ANALYSIS REPORT

Wave 5.2R Stage 6 Spectral And Sobolev Guidance Results

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Decision

Stage 6 is complete as a valid negative result.

All `15` first-screen candidates completed without runtime failures, but no eligible formulation passed the complete multi-index gate. The conditional stability continuation was therefore correctly not started.

`FI01`, which combines bounded H04 coefficients, derivative and spectral guidance, and training-only failure-informed angular weights, is the raw-error leader at `0.001710638 deg`. This is a `0.88%` improvement over the frozen Stage 5 H04 seed. Nevertheless, FI01 slightly worsens Sobolev derivative MAE by `0.012%` and reduces derivative correlation by `0.002435`. It is not promoted.

Stage 5 H04 remains the qualified structured component entering Stage 7. No Stage 6 model replaces the accepted periodic GRU or becomes a production candidate.

Scope And Integrity

- dataset: `polished_dataset`;
- input contract: setpoints only;
- surface: `Fw`;
- accepted curves: `966`;
- split: `675` train, `194` validation, `97` test;
- angular grid: `2048` uniform points;
- split signature: `c1aa8718fb9bf88cc2021c121dc4f3b4010fc1d2e45ac90af5f4376aa64f8e16`;
- first-screen seed: `314159`;
- target-derived runtime inputs: none;
- failed runs: `0`;
- official TE Curve Verification Pipeline: not run.

  The preflight passed every derivative, spectrum, coordinate-bound, model-shape, and leakage check. The second-derivative sensitivity gate failed, so curvature supervision remained disabled before training as designed.

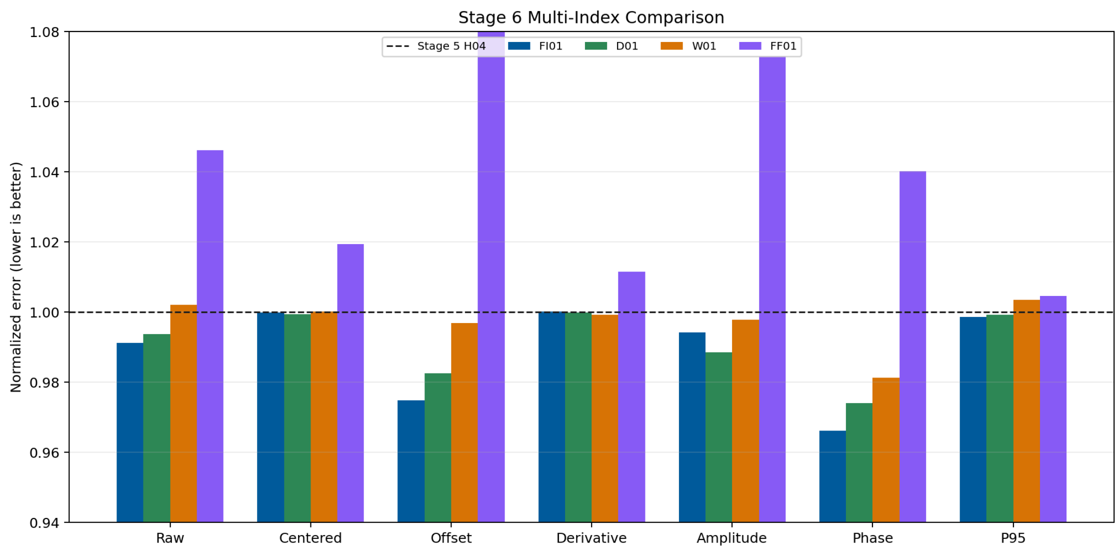
Campaign Matrix

Family	Candidates	Scientific question
Coefficient controls	C01, C02, C03, C04	Does representation or direct coefficient prediction explain the gain?
Sobolev and spectral	D01, S02, DS01, DS02	Do first derivatives or complex frequency targets add held-out value?

Family	Candidates	Scientific question
Training strategies	CU01, FI01	Does curriculum or localized failure weighting help?
Coordinate networks	FF00, FF01, SI00, SI01	Do Fourier features or SIREN resolve missed angular structure?
Weak form	W01	Do local Fourier moments help without pointwise derivative noise?

Primary Results

Candidate	Raw MAE [deg]	Centered [deg]	Offset [deg]	D-MAE	D-corr.	P95 [deg]
Stage 5 H04	0.0017259	0.0013555	0.0008844	0.2810657	0.3407822	0.0039334
FI01	0.0017106	0.0013554	0.0008622	0.2811006	0.3383469	0.0039280
D01	0.0017152	0.0013546	0.0008690	0.2810336	0.3398775	0.0039307
W01	0.0017294	0.0013557	0.0008817	0.2808554	0.3411658	0.0039473
FF01	0.0018055	0.0013819	0.0009719	0.2842875	0.3250251	0.0039516



Stage 6 multi-index comparison

Gate Matrix

Candidate	Raw	Centered	Offset	D-MAE	D-corr.	Amp.	Phase	P95	Ctrl.	Final
D01	pass	pass	pass	fail	fail	pass	pass	pass	pass	fail
S02	fail	pass	fail	fail	pass	fail	fail	pass	pass	fail
DS01	pass	pass	pass	fail	fail	pass	pass	pass	pass	fail
DS02	fail	pass	fail	fail	pass	fail	fail	pass	pass	fail
CU01	pass	pass	pass	fail	fail	fail	pass	pass	pass	fail
FI01	pass	pass	pass	fail	fail	pass	pass	pass	pass	fail
FF01	fail	fail	fail	fail	fail	fail	fail	fail	pass	fail
SI01	fail	fail	fail	fail	fail	fail	fail	fail	fail	fail
W01	pass	pass	pass	fail	pass	fail	pass	fail	fail	fail

No row passes all gates.

What Worked

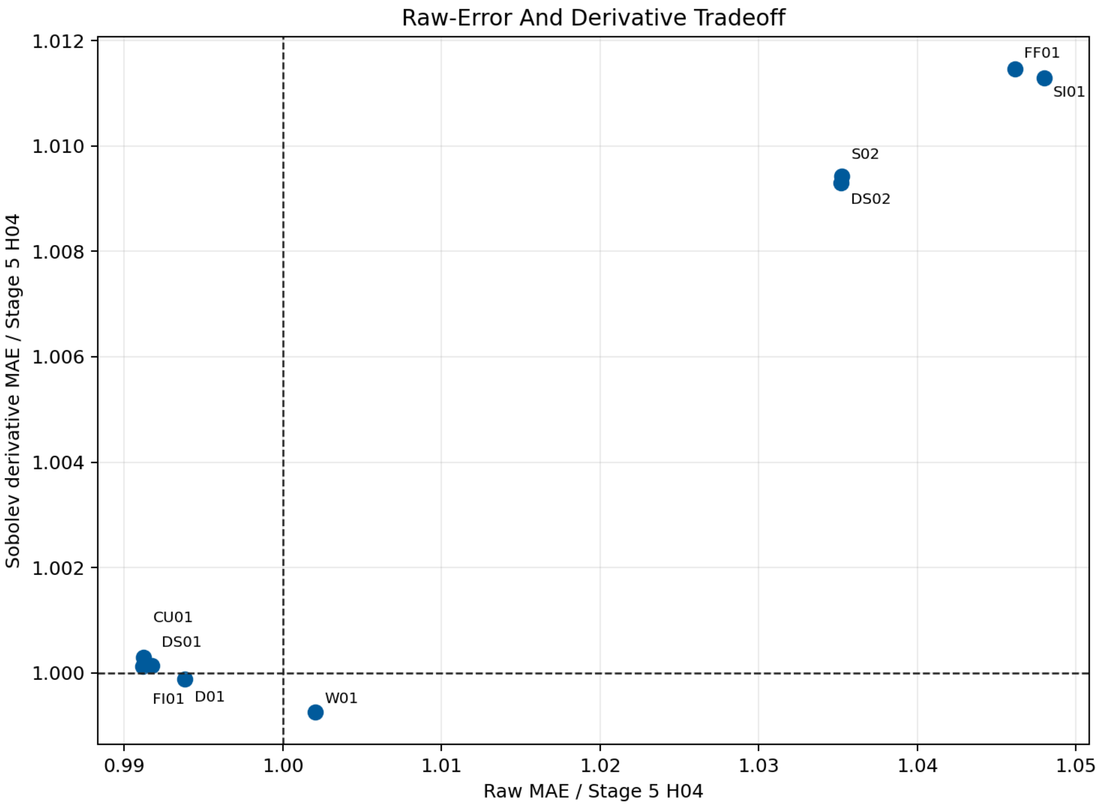
- FI01, CU01, DS01, and C02 all reduce raw MAE relative to Stage 5 H04.
- FI01 preserves centered shape, offset, amplitude, phase, and P95 within the declared gates while beating its DS01 matched control.
- D01 obtains the best centered-shape result among the leading bounded coefficient candidates and improves harmonic amplitude and phase.
- W01 produces the best derivative MAE and derivative correlation in the first screen, showing that weak local moments can alter the intended differential behavior.
- Every model keeps unsupported high-frequency energy bounded.
- The preflight successfully rejected unstable second-derivative supervision.

What Did Not Work

- The pointwise derivative candidates fail the required derivative improvement thresholds despite their explicit Sobolev losses.
- FI01's raw-error gain does not transfer to derivative correlation.
- W01's derivative gain costs raw error, amplitude, and tail quality and does not beat its C01 control.
- Fragile-band H08 formulations S02 and DS02 worsen raw error and offset.
- Direct coefficient candidates C03 and C04 remain substantially worse.
- Fourier-feature and SIREN coordinate residuals do not beat the bounded coefficient family and do not demonstrate a useful spectral-bias correction.
- No candidate earns stability continuation or model promotion.

Raw-Error And Derivative Tradeoff

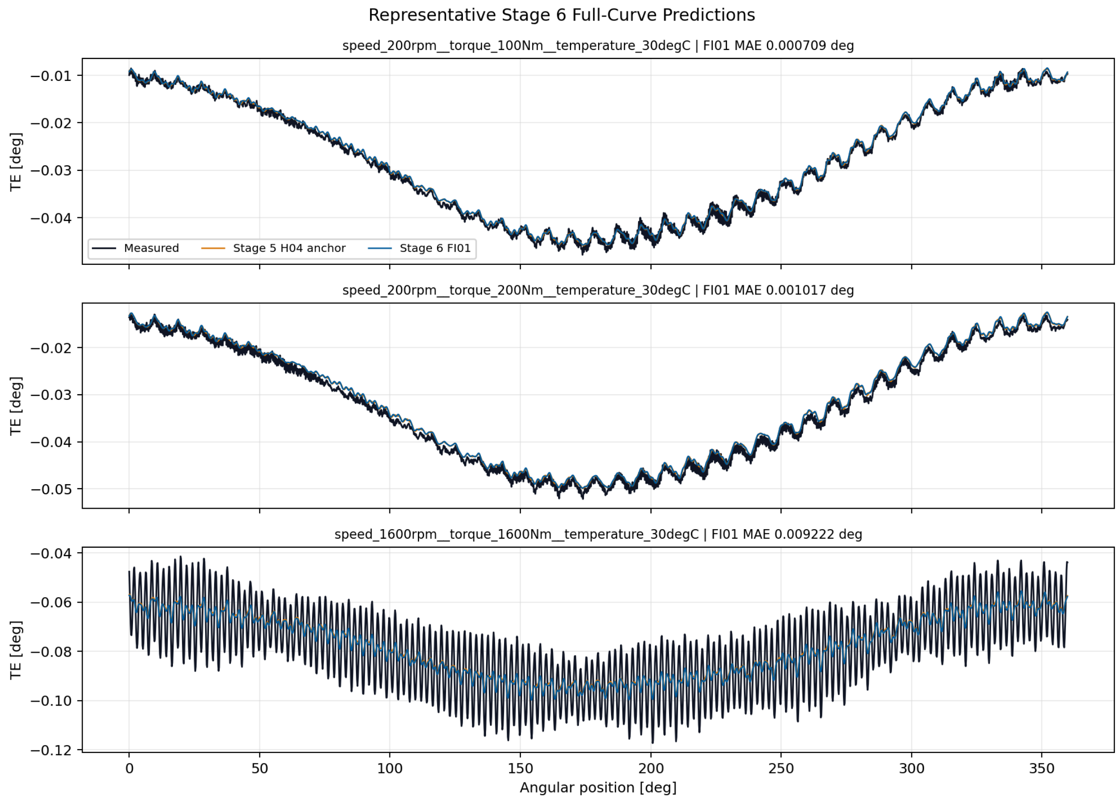
The lower-left quadrant would improve both quantities relative to Stage 5 H04. No candidate reaches the required improvement region with the remaining gates.



Raw-error and derivative tradeoff

Representative Full Curves

FI01 remains visually close to the qualified H04 component. Its improvement is small and distributed; it does not expose a new localized correction capable of satisfying the derivative gate.



Representative FI01 curves

Scientific Interpretation

Stage 6 does not show that spectral or Sobolev guidance is useless. It shows that, on the current bounded coefficient representation and fixed split, the tested loss formulations mostly redistribute error among already correlated curve metrics. A raw-curve gain can coexist with a worse derivative field, and a weak-form derivative gain can coexist with worse tails or harmonic amplitude.

This is exactly why the multi-index gate is necessary. Selecting FI01 by MAE alone would overstate the physics contribution.

The next controlled hypothesis is decomposition rather than another global loss mixture: Stage 7 separates the mean/offset quantity from the mean-centered periodic shape. That design directly targets the offset-shape competition visible here while preserving H04 as an inspectable structured component.

Program Decision

- Stage 6 status: complete, valid negative result;
- completed runs: `15 / 15`;
- stability runs: `0`, correctly skipped;
- raw-error leader: FI01;
- promoted Stage 6 candidate: none;
- retained component: Stage 5 H04;
- production or registry promotion: no;
- next step: Stage 7, Mean And Centered-Shape Multi-Head Model.

Artifact Map

- campaign: `output/training_campaigns/2026-07-29-15-34-05_wave52r_stage6_spectral_sobolev_guidance_2026_07_29/`;
- gate summary: `output/analysis/wave_5_2r/stage6_spectral_sobolev_guidance/closeout/stage6_exit_gate_summary.yaml`;
- preflight: `output/analysis/wave_5_2r/stage6_spectral_sobolev_guidance/stage6_preflight_validation_summary.yaml`;
- FI01 checkpoint: `output/training_runs/spectral_sobolev_guidance/2026-07-29-15-34-13__stage6_fi01/best_model.pt`;
- model report: `doc/reports/analysis/model_development_waves/wave_5_2/physics_guided_pinn_reassessment/[2026-07-29]/stage6_spectral_sobolev_guidance/stage6_spectral_sobolev_guided_residual_model_report.md`.